MATH 1564 — Notes 09/01

Homogeneous systems

Definition: $Ax = 0$ is called homogeneous.

Key property: For any number c, if $Ax = 0$, then $A(cx) = 0$.

Why? Write $A = [a_1 \ \cdots \ a_n]$ and $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$. Then

$$A(cx) = cx_1a_1 + \cdots + cx_na_n = c(x_1a_1 + \cdots + x_na_n) = cAx = c0 = 0.$$

Example: $Ax = 0$ always has the obvious solution $x = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$.

Theorem 1: The following are equivalent:

1. $Ax = 0$ has a nonzero solution.
2. $Ax = 0$ has infinitely many solutions.
3. There is a free variable.
4. A has a column with no pivot.

Convention: Since columns do not move under row operations, we say a column of A is **pivotal** if the same column in RREF(A) has a pivot.

$2 \iff 3$: trivial.

$1 \Rightarrow 2$: If x is a nonzero solution of $Ax = 0$, then cx is also a solution for any number c, and these are all distinct since $x \neq 0$ — so there are infinitely many solutions.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \to \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

No pivot in column 2. A nonzero solution of $Ax = 0$ is $x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, so there are infinitely many solutions.

Linear dependence

Motivation: If $v_1, \dots, v_k$ are vectors in $\mathbb{R}^m$, how do we describe $\operatorname{span}\{v_1, \dots, v_k\}$ with the least amount of data?

Definition: Vectors $\{v_1, \dots, v_k\}$ in $\mathbb{R}^m$ are **linearly independent** if the vector equation $c_1v_1 + \cdots + c_kv_k = 0$ with unknowns $c_1, \dots, c_k$ only has the zero solution

$$\begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Definition: Vectors $\{v_1, \dots, v_k\}$ are **dependent** if they are not independent.

Theorem 2: $\{v_1, \dots, v_k\}$ is dependent $\iff$ for some $i = 1, \dots, k$, v_i is a linear combination of $v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_k$.

Proof: We have $c_1v_1 + \cdots + c_kv_k = 0$ for some $c_i \neq 0$. Solve for v_i:

$$v_i = -\frac{1}{c_i}\left(c_1v_1 + \cdots + c_{i-1}v_{i-1} + c_{i+1}v_{i+1} + \cdots + c_kv_k\right).$$

Conversely, if v_i equals such a combination, moving it to the other side gives a nontrivial relation $c_1v_1 + \cdots + c_kv_k = 0$.

Example: Find all h such that the columns of $\begin{bmatrix} 1 & 2 \\ 3 & h \end{bmatrix}$ are dependent.

$h = 6$: $\begin{bmatrix} 2 \\ 6 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

Note: $\{v_1, v_2\}$ is dependent $\iff$ $v_1 = cv_2$ or $v_2 = cv_1$ for some c.

Example: Pivotal columns of a RREF matrix are independent.

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Example: If some $v_i = 0$, then $v_1, \ldots, v_k$ is dependent.

Take $i = 1$: $1 \cdot v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_k = 0$ is a nontrivial relation.

Theorem 3: Let $A = [v_1 \ \cdots \ v_n]$ be an $m \times n$ matrix, where v_i is column i of A. Then

$\{v_1, \ldots, v_n\}$ is independent $\iff$ each column of A is pivotal (RREF has n pivots).

Proof: $c_1 v_1 + \cdots + c_n v_n = 0 \iff A \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = 0.$

A homogeneous system has only the zero solution $\iff$ each column of A is pivotal.

More general.

Theorem 4: The pivotal columns of any matrix are independent.

Proof: Suppose $A = [v_1 \ \cdots \ v_n]$ and $v_{i_1}, \ldots, v_{i_k}$ are the pivotal columns.

Let $c = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ with $c_i = 0$ if i is not among $i_1, \ldots, i_k$.

Suppose $Ac = c_{i_1} v_{i_1} + \cdots + c_{i_k} v_{i_k} = 0$; we show $c = 0$.

Now $Ac = 0 \iff Rc = 0$, where $\mathrm{RREF}(A) = R = [r_1 \ \cdots \ r_n]$. So

$$0 = Rc = c_1 r_1 + \cdots + c_n r_n = c_{i_1} r_{i_1} + \cdots + c_{i_k} r_{i_k}.$$

Since A has k pivots and $r_{i_1}, \ldots, r_{i_k}$ are exactly the pivot columns of R, they are the standard basis vectors of $\mathbb{R}^k$ (in their nonzero rows), so this equation forces $c_{i_1} = \cdots = c_{i_k} = 0$ by matching components, i.e., $c = 0$. Hence $\{v_{i_1}, \ldots, v_{i_k}\}$ is independent.

Conclusion: To test if vectors are independent, put them as columns in a matrix and check if all columns are pivotal.

Example: Find all h such that $\begin{bmatrix}1\\1\\h\end{bmatrix}, \begin{bmatrix}1\\h\\1\end{bmatrix}, \begin{bmatrix}h\\1\\1\end{bmatrix}$ are independent.

$$\begin{bmatrix}1 & 1 & h\\1 & h & 1\\h & 1 & 1\end{bmatrix} \equiv \begin{bmatrix}1 & 1 & h\\0 & h-1 & 1-h\\0 & 1-h & 1-h^2\end{bmatrix} \equiv \begin{bmatrix}1 & 1 & h\\0 & h-1 & 1-h\\0 & 0 & 2-h-h^2\end{bmatrix}$$

Need $h-1 \neq 0$ and $2-h-h^2 \neq 0$ (if either $=0$, the columns are dependent).

$$2-h-h^2 = -(h-1)(h+2) = 0 \iff h = 1 \text{ or } h = -2.$$

So we need $h \neq 1$ and $h \neq -2$.

In that case the matrix reduces to $\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$, so the vectors are independent.

Example: If v is in $\mathbb{R}^n$, is $\{v\}$ dependent?

$cv = 0 \Rightarrow c = 0$ is true $\iff v \neq 0$. So $\{v\}$ is independent iff $v \neq 0$.

Theorem 5: If $v_1, \dots, v_k$ are in $\mathbb{R}^n$ and $k > n$, then $\{v_1, \dots, v_k\}$ is dependent.

Proof: Let $A = [v_1 \ \cdots \ v_k]$, an $n \times k$ matrix. Suppose all columns are pivotal; then there are k pivots. But there are only n rows and each pivot lies in its own row, so the number of pivots is at most n. Since $k > n$, this is a contradiction, so the columns are not all pivotal. Therefore $\{v_1, \dots, v_k\}$ is dependent.

Example: Suppose v_1, v_2, v_3 are in $\mathbb{R}^2$:

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_1 + v_2 = v_3.$$

$\{v_1, v_2\}$, $\{v_1, v_3\}$, $\{v_2, v_3\}$ are each independent, but $\{v_1, v_2, v_3\}$ is not independent (by Theorem 5, since $k = 3 > 2 = n$).